

TERRA NOVA NATL PARK, NL, Canada

WMO: 715890

Lat: 48.560N

Lon: 53.970W

Elev: 107

StdP: 100.05

Time Zone: -3.50 (NAF)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-16.9	-14.8	-21.5	0.6	-15.7	-19.2	0.7	-13.6	11.4	-3.4	10.4	-3.3	3.9	250	0.659

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.6	27.5	19.0	25.6	18.3	23.8	17.7	20.9	24.6	19.8	23.3	18.8	22.0	4.6	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.6	14.5	22.5	18.6	13.6	21.6	17.5	12.7	20.4	60.9	24.6	57.2	23.3	53.8	22.2	25.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
9.2	8.1	7.2	DB	-19.8	31.1	2.3	1.3	-21.5	32.0	-22.9	32.7	-24.2	33.4	-25.9	34.4	
			WB	-20.0	22.8	2.2	1.1	-21.6	23.6	-22.9	24.2	-24.2	24.8	-25.8	25.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.3	-5.4	-5.4	-2.6	2.2	6.9	11.8	17.1	17.1	13.1	7.5	2.7	-2.4		
	DBStd	8.88	4.60	4.61	3.92	3.47	3.80	4.33	3.71	3.16	3.88	3.81	3.95	4.50		
	HDD10.0	2393	476	431	391	235	113	29	2	1	13	98	221	383		
	HDD18.3	4822	734	664	649	484	355	200	66	60	162	336	468	642		
	CDD10.0	672	0	0	0	1	16	83	223	222	106	20	2	0		
	CDD18.3	61	0	0	0	0	0	4	29	23	5	0	0	0		
	CDH23.3	476	0	0	0	0	6	59	236	139	36	0	0	0		
	CDH26.7	84	0	0	0	0	1	9	48	18	8	0	0	0		
(14) Wind	WSAvg	3.5	4.0	4.0	3.8	3.5	3.2	3.2	3.2	3.1	3.4	3.4	3.6	3.8		
(15) Precipitation	PrecAvg	1261	110	109	107	88	91	90	97	102	111	125	119	125		
	PrecMax	1492	215	146	217	164	173	165	184	159	203	220	248	251		
	PrecMin	970	52	35	55	42	36	32	45	44	44	59	35	31		
	PrecStd	122	38	24	37	32	32	35	33	28	43	46	52	47		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.3	8.1	9.4	15.7	23.0	27.9	30.1	28.9	27.4	20.0	15.1	11.4		
		MCWB	9.9	7.2	6.2	9.3	13.6	18.5	20.1	19.4	19.9	16.1	13.4	11.2		
	2%	DB	5.7	5.3	6.3	12.6	19.4	25.0	28.0	26.5	23.6	17.7	13.3	8.8		
		MCWB	5.2	4.4	4.1	8.8	12.3	16.9	19.4	18.7	18.0	15.4	12.4	8.4		
	5%	DB	2.9	2.8	4.4	10.4	16.6	22.3	26.1	24.8	21.2	15.7	11.5	6.1		
		MCWB	2.5	1.8	2.2	7.0	11.1	15.6	18.7	18.3	16.7	13.4	10.7	5.5		
	10%	DB	0.9	0.9	2.7	8.0	13.9	19.6	24.0	23.0	19.4	13.8	9.2	3.7		
		MCWB	0.4	0.0	1.0	5.4	9.9	14.3	17.9	17.7	15.9	11.8	8.3	3.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.9	7.5	7.4	11.8	16.1	20.9	22.4	22.1	21.1	18.0	14.1	11.3		
		MCDB	10.2	7.9	8.5	13.1	20.4	26.0	27.3	25.7	24.9	18.8	14.6	11.4		
	2%	WB	5.4	4.5	4.6	9.5	13.9	18.5	21.0	20.7	19.3	16.0	12.5	8.4		
		MCDB	5.6	5.1	5.6	11.6	17.1	22.5	25.3	23.6	22.0	17.0	13.2	8.7		
	5%	WB	2.5	2.0	2.7	7.5	11.9	16.5	19.9	19.6	18.0	13.9	10.7	5.7		
		MCDB	2.8	2.7	3.9	9.7	15.6	20.6	24.1	22.6	20.4	15.3	11.4	6.0		
	10%	WB	0.5	0.2	1.2	5.6	10.1	14.8	18.8	18.6	16.6	11.9	8.4	3.3		
		MCDB	0.8	0.7	2.2	7.5	13.4	18.9	22.3	21.8	18.8	13.4	9.1	3.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.7	8.4	8.0	8.6	9.9	10.6	10.7	9.6	9.6	7.7	6.8	6.4		
		MCDBR	9.5	10.2	9.9	12.4	15.2	15.4	13.9	12.5	12.0	10.1	9.5	9.1		
	5% WB	MCWBR	9.6	9.7	7.8	8.4	9.1	8.3	6.7	6.4	7.3	7.6	9.1	9.2		
		MCWBR	9.6	9.9	9.0	11.2	13.4	13.4	11.7	10.5	10.7	9.3	9.4	9.1		
(40) Clear-Sky Solar Irradiance	taub		0.308	0.303	0.336	0.359	0.372	0.398	0.416	0.396	0.361	0.339	0.314	0.304		
	taud		2.273	2.354	2.312	2.362	2.394	2.344	2.343	2.396	2.480	2.502	2.490	2.330		
	Ebn at Noon		740	853	876	890	889	864	842	843	843	805	750	697		
	Edh at Noon		75	86	106	113	114	120	119	108	90	75	61	63		
(44) All-Sky Solar Radiation	RadAvg		0.97	1.77	2.89	4.02	4.73	5.07	5.20	4.34	3.36	2.02	1.13	0.73		
	RadStd		0.09	0.17	0.26	0.42	0.37	0.48	0.46	0.31	0.23	0.21	0.14	0.07		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page